PROJECT REPORT

on

LCD Display and Matrix Key Interface Upgrade for Ultrasonic Thickness Gauge

Submitted in partial fulfilment of the requirements for the degree of

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CERTIFICATE

This is to certify that the Project Report entitled " LCD Display and Matrix Key Interface Upgrade for Ultrasonic Thickness Gauge” has been carried out by Dwij Vyas (College ID No: 24EC436) under the guidance of **Modsonic Instruments Mfg. Co.(P) Ltd, Anand,** in partial fulfilment of the requirement for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering from Birla Vishvakarma Mahavidyalaya (BVM), Vallabh Vidyanagar, during the academic year 2025–2026.

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ABSTRACT

This report documents the design, implementation, and testing of two embedded systems projects completed during an industrial internship at Modsonic Instruments Mfg. Co.(P) Ltd, Anand, Gujarat.

Project 1 — TFT LCD Display and Capacitive Keypad Interface Upgrade for Ultrasonic Thickness Gauge: The legacy Ultrasonic Thickness Gauge Ultrasonic Thickness Gauge (UTG) relied on a 4-digit segment LCD and four mechanical push buttons for its user interface. This project replaced that interface entirely with a 320×240 IPS TFT colour display (ST7789VI) driven over SPI at 12 MHz, and an 8-key capacitive touch keypad (TTP229) read via a bit-bang serial protocol, both interfaced to the AVR128DA48 microcontroller at 24 MHz. The complete measurement pipeline — ultrasonic pulse-echo timing, 12-bit counter accumulation, calibration algorithms, EEPROM persistence, and power management — was preserved intact. A professional instrument user interface was designed from scratch, featuring a top bar with the Modsonic logo, MM/INCH indicator, and live battery percentage; a large-digit measurement content area; a hint bar for key labels; and a full hidden diagnostic mode (brightness control, button test, volume, and system information). The project required systematic counter bus remapping across 12 signals to resolve pin conflicts introduced by the TFT and TTP229.

Project 2 — 4-Digit 7-Segment LCD Display Driver using PCF85176 IC: This project involved the complete firmware development to drive a Clover Display S5026 4-digit 7-segment passive LCD panel using the PCF85176 40-segment I²C LCD driver IC, controlled by an AVR128DA48 Curiosity Nano development board. The PCF85176 was configured in static drive mode via a custom I²C driver using TWI0 on PA2/PA3. The firmware implements a flexible segment-pin definition architecture allowing any S-pin (S0–S39) to be freely mapped to any digit segment, catering to arbitrary PCB routing requirements. A complete segment lookup table, RAM-based display buffer, and byte-to-digit rendering function were implemented to display arbitrary 7-segment patterns on all four digits.

Both projects were developed in C for the AVR128DA48 using MPLAB X IDE and the XC8 compiler.

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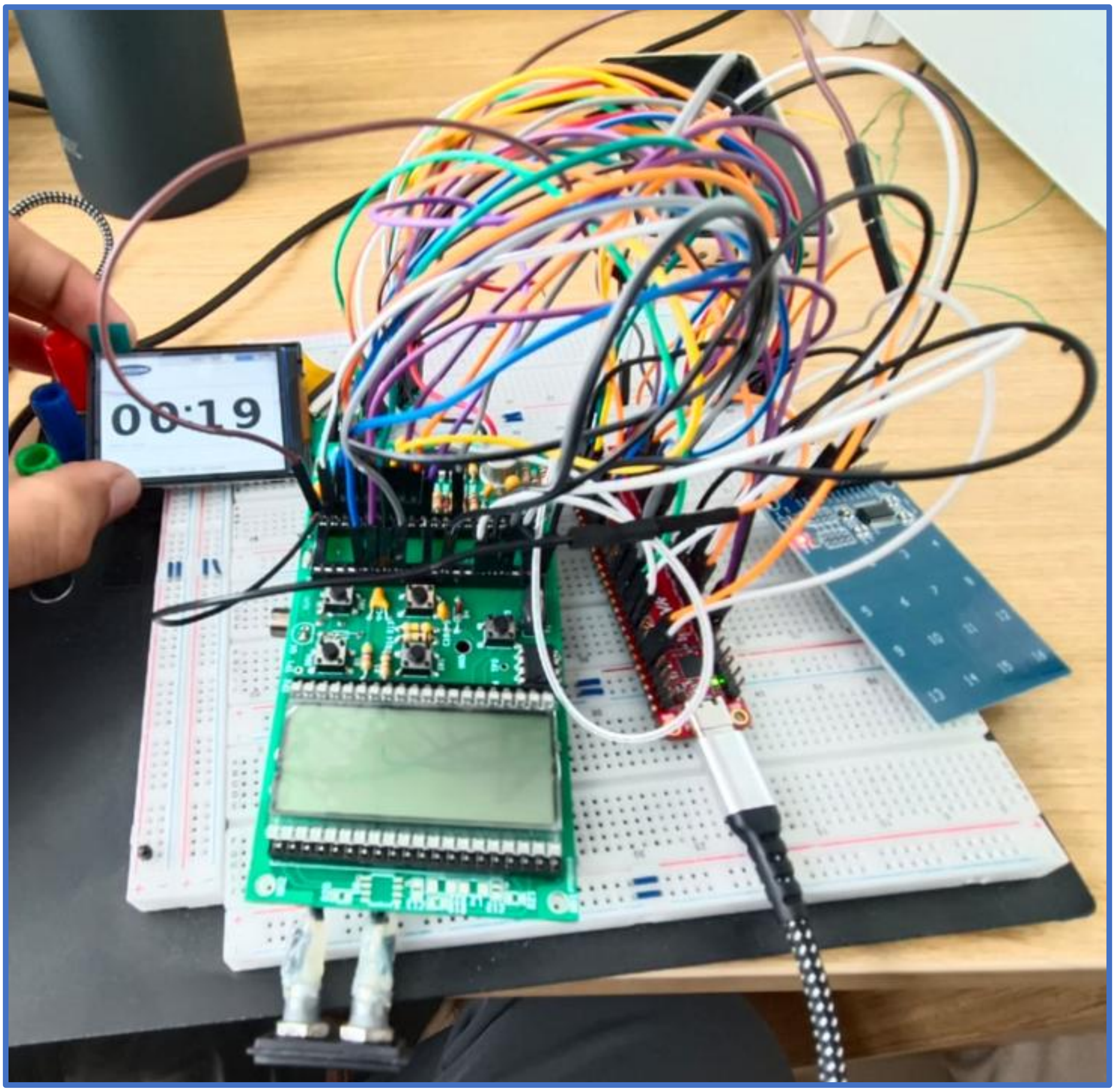
List of Abbreviations

Abbreviation	Full Form
AC	Alternating Current
AVR	Alf and Vegard's RISC (processor family)
CABC	Content Adaptive Backlight Control
COM	Common (LCD backplane electrode)
DC	Direct Current / Data-Command (TFT control signal)
EEPROM	Electrically Erasable Programmable Read-Only Memory
FPC	Flexible Printed Circuit
GPIO	General Purpose Input/Output
I²C	Inter-Integrated Circuit (serial bus protocol)
IDE	Integrated Development Environment
IPS	In-Plane Switching (LCD technology)
ISR	Interrupt Service Routine
LCD	Liquid Crystal Display
MCU	Microcontroller Unit
MIPS	Million Instructions Per Second
MOSI	Master Out Slave In (SPI data line)
NDT	Non-Destructive Testing
NVM	Non-Volatile Memory
PCB	Printed Circuit Board
PZT	Lead Zirconate Titanate (piezoelectric material)
RAM	Random Access Memory
RGB565	16-bit colour format: 5 bits Red, 6 bits Green, 5 bits Blue
RISC	Reduced Instruction Set Computer
SCK	Serial Clock (SPI)
SCL	Serial Clock Line (I²C / TTP229)
SDA	Serial Data Line (I²C)
SDO	Serial Data Output (TTP229)
SPI	Serial Peripheral Interface
SRAM	Static Random Access Memory
TFT	Thin-Film Transistor (LCD type)
TWI	Two-Wire Interface (Atmel/Microchip term for I²C)

PROJECT 1

TFT LCD Display and Capacitive Keypad Interface Upgrade for Ultrasonic



Thickness Gauge

Chapter 1: Introduction

1.1 About Modsonic Instruments Mfg. Co.(P) Ltd

Modsonic Instruments is a leading Indian manufacturer of Non-Destructive Testing (NDT) equipment, headquartered in Anand, Gujarat. Established with a focus on ultrasonic testing technology, the company designs, manufactures, and markets a range of portable testing instruments used extensively in manufacturing, power generation, oil and gas, aerospace, and infrastructure maintenance industries. Their product portfolio includes ultrasonic thickness gauges, flaw detectors, and related calibration accessories.

The company serves both domestic and international markets and maintains an in-house R&D team responsible for firmware development, hardware design, and product innovation. The instruments are designed for field use in harsh industrial environments, requiring robust hardware, low power consumption, and reliable long-term operation.

1.2 Project Background and Motivation

The Ultrasonic Thickness Gauge is Modsonic's production Ultrasonic Thickness Gauge (UTG) — a portable, battery-operated handheld instrument designed for industrial field measurements. The original Ultrasonic Thickness Gauge device used a 4-digit segment (clover) LCD display driven by the MM5453 serial LCD driver chip, and four mechanical push buttons as its user interface. While this interface was functionally adequate, it imposed several practical limitations: limited display real estate confined the readout to four numeric digits with no contextual information, mechanical buttons were subject to wear over extended field use, and the monochrome segment display offered no possibility of colour-coded status indicators, graphical elements, or future UI expansion.

This internship project was assigned to upgrade the Ultrasonic Thickness Gauge user interface to a modern 320×240 IPS TFT colour display (ST7789VI) and an 8-key capacitive touch keypad (TTP229), while keeping all original measurement functionality — ultrasonic thickness measurement, calibration routines, EEPROM storage, and power management — completely intact and unmodified in their logic.

1.3 Objectives of the Project

- Replace the 4-digit segment LCD with a 320×240 IPS TFT colour display (ST7789VI) controlled via SPI.
- Replace the 4 mechanical push buttons with an 8-key TTP229 capacitive touch keypad.

- Design and implement a professional, production-grade instrument user interface on the TFT display.
- Preserve all original measurement algorithms, calibration functions, EEPROM structures, and power management logic without modification.
- Resolve all hardware pin conflicts between the new peripherals and the existing counter bus and push button pins through systematic remapping.
- Implement a hidden diagnostic mode accessible via Key 8 for factory testing and field service.
- Document all hardware connections, firmware modules, and design decisions comprehensively.

1.4 Scope

This project covers complete hardware pin planning, SPI and bit-bang serial driver development, TFT display driver implementation, instrument UI design, capacitive keypad integration, counter bus remapping, and diagnostic HMI development. The project was implemented in C for the AVR128DA48 microcontroller using MPLAB X IDE and the XC8 compiler. The ultrasonic measurement pipeline (CYCLE, READ_CNT, SUBDELTA, DIR_CAL, calibration functions) remains unchanged from the existing firmware.

Chapter 2: System Overview and Components

2.1 Original Hardware Platform

The Ultrasonic Thickness Gauge is built around the AVR128DA48 microcontroller running at 24 MHz. In its original configuration, the device interfaced with the following peripherals:

- Display: 4-digit segment (clover) LCD, driven via PC0 (clock) and PC1 (data) using a 2-wire serial bit-bang protocol to the MM5453 LCD driver IC.
- Keypad: 4 mechanical push buttons on PA1 (PROBE), PA2 (DOWN), PA3 (CAL), PA4 (UP) with active-low inputs and internal pull-ups.
- Power Button: PA0 — falling-edge interrupt for clean shutdown.
- Power Latch: PA5 — held HIGH by firmware to keep the device powered on.
- Low Battery Detect: PA6 — active-low input.
- Counter Bus: 12-bit ripple counter outputs on PB0–PB5 (Q1–Q6), PD0–PD2 (Q7–Q9), PC4–PC6 (Q10–Q12).
- Measurement Signals: MCLOCK=PD3 (master clock to ultrasonic pulser), COU_R=PD4 (counter reset), CLK_10=PC7 (10 MHz oscillator enable), ECHO=PF0 (transducer receive circuit).

2.2 New Components Added

Two new peripheral ICs are introduced in this upgrade:

TFT LCD Display : This is a single-chip IPS TFT LCD controller embedded in the 30-pin FPC connector of a 2.8" 320×240 colour LCD module. It communicates over a 4-wire SPI interface at up to 16 MHz and supports 16-bit RGB565 colour pixel data. The ST7789VI requires a hardware reset sequence at power-on followed by a specific register initialisation sequence covering MADCTL (memory access control), pixel format, porch settings, gate control, VCOMS, LCM control, VDV and VRH settings, framing rate, power control, and positive/negative gamma curves.

TTP229 Capacitive Touch Keypad Controller: The TTP229 is an 8/16-key capacitive touch controller IC from Tontek. In its 8-key serial output mode, the host generates a clock on SCL and reads 8 bits of key state on SDO, one bit per clock cycle, LSB first. Each bit is '0' when the corresponding key is touched and '1' when released (active-low). The TTP229 module used provides 8 capacitive sensing pads (TP1–TP8).

2.3 Component Summary Table

Component	Part Number	Interface	Purpose
MCU	AVR128DA48	N/A	Main microcontroller, 24 MHz
TFT LCD Display	ST7789VI module	SPI (12 MHz)	320×240 colour UI display
Capacitive Keypad	TTP229 module	Serial bit-bang	8-key touch input
Memory	Internal EEPROM	NVMCTRL	Calibration persistence
Counter IC	74HC4040	Parallel GPIO (12-bit)	Ultrasonic transit time
Battery	9V / 3.7V Li	Analog PA6	Power supply

Table 2.1: Component Summary — Project 1 (Ultrasonic Thickness Gauge TFT Upgrade)

Chapter 3: Hardware Architecture and Pin Connections

3.1 LCD TFT Display (SPI Interface)

The ST7789VI communicates via 4-wire SPI. The AVR128DA48 SPI peripheral (SPI0) is used in master mode with CLK2X enabled and PRESC_DIV4 prescaler, yielding 12 MHz SPI clock from the 24 MHz system clock. The PORTMUX is left at default so SPI0 maps to PA4 (MOSI), PA6 (SCK), PA5 (MISO unused), and PA7 (SS unused). Three additional control signals — DC (Data/Command), RST (Reset), and CS (Chip Select) — are driven as GPIO outputs on PD0, PD1, and PD2 respectively.

Signal	AVR128DA48 Pin	Notes
MOSI / SDA	PA4	SPI data output
SCK / SCL	PA6	SPI clock output
DC / RS	PD0	LOW = command, HIGH = data
RST / RES	PD1	Active LOW hardware reset
CS	PD2	Active LOW chip select
VDD	3.3V rail	Logic and LCD supply
GND	GND	Ground
LED_A (BL+)	3.3V rail	Backlight anode
LED_K1–K4 (BL−)	GND	Backlight cathodes

Table 3.1: ST7789VI TFT Pin Connections to AVR128DA48

3.2 TFT 30-Pin FPC Connector Wiring

The TFT Display module uses a 30-pin FPC (Flexible Printed Circuit) connector. The interface mode is selected by the IM0–IM3 pins. For 4-wire SPI mode, IM0 is tied LOW and IM1–IM3 are tied HIGH. The parallel data pins D0–D7 are left open as they are unused in SPI mode. The complete pin assignment is as follows:

FPC Pin	Signal	Connection
1	VSS	GND
2	NC	Leave open
3	NC/SDO	Leave open
4	TE	Leave open
5–12	D0–D7	Leave open (parallel, unused in SPI)

FPC Pin	Signal	Connection
13	SDA (MOSI)	PA4
14	/RD	3.3V (tie HIGH)
15	/WR / DC	PD0
16	D/C / SCL	PA6 (SCK)
17	/CS	PD2
18	/RES	PD1
19	IM0	GND (SPI mode select)
20	IM1	3.3V
21	IM2	3.3V
22	IM3	3.3V
23	VDD	3.3V
24	VSS	GND
25	LED_A	3.3V (backlight anode)
26–29	LED_K1–K4	GND (backlight cathodes)
30	VSS	GND

Table 3.2: TFT 30-pin FPC Connector — Pin-by-Pin Connections

3.3 TTP229 Capacitive Keypad

The TTP229 is connected to the AVR128DA48 using two GPIO pins. PB1 is configured as an output and drives the SCL clock line. PB0 is configured as an input with an internal pull-up and reads the SDO data line. During a read cycle, the AVR generates 8 falling clock edges on PB1. On each falling edge, the TTP229 shifts one bit of key state onto SDO. The bit is sampled by the AVR after the falling edge. This bit-bang serial read is performed in the ttp_read() function.

Signal	AVR128DA48 Pin	Notes
VCC	3.3V	Power supply
GND	GND	Ground
SCL	PB1	Clock output from AVR — idle HIGH

Signal	AVR128DA48 Pin	Notes
SDO	PB0	Serial data input to AVR, active LOW
TP1–TP8	Touch pads	8 capacitive sensing electrodes

Table 3.3: TTP229 Capacitive Keypad Pin Connections

3.4 Key Function Mapping

The 8 capacitive keys of the TTP229 replace and extend the original 4 mechanical push buttons. Each key corresponds to a bit in the 8-bit value returned by ttp_read(). Key bits are active-HIGH in firmware (after inversion of the TTP229's active-LOW output). The mapping is as follows:

TTP229 Key	Bit Mask	Replaces	Function
K1 (TP1)	0x01	PA1 (PROBE key)	Measure / Zero calibration
K2 (TP2)	0x02	PA2 (DOWN key)	Decrease value / Confirm
K3 (TP3)	0x04	UP key (hold at boot)	MM / INCH unit toggle
K4 (TP4)	0x08	—	Reserved
K5 (TP5)	0x10	PA1 (PROBE)	PROBE — zero calibration
K6 (TP6)	0x20	PA3 (CAL key)	Known-thickness calibration / Back
K7 (TP7)	0x40	—	Reserved
K8 (TP8)	0x80	New function	Diagnostic mode toggle

Table 3.4: TTP229 to AVR Key Mapping (8-key)

Chapter 4: Counter Bus Remapping

4.1 Pin Conflict Analysis

The introduction of the TFT LCD Display and TTP229 capacitive keypad created direct conflicts with the existing counter bus and push button pin assignments:

- PB0 and PB1: Originally used for counter outputs Q1 and Q2, these pins are now required by the TTP229 (SDO=PB0, SCL=PB1). They cannot be shared.
- PD0, PD1, and PD2: Originally used for counter outputs Q7, Q8, and Q9, these pins are now required by the TFT display (DC=PD0, RST=PD1, CS=PD2). They cannot be shared.
- PA1, PA2, PA3, PA4: Originally used for the 4 mechanical push buttons. Since the TTP229 replaces the push buttons entirely, these pins are freed and available for other functions.

The resolution strategy was to remap all 12 counter bus signals (Q1–Q12) to a set of free GPIO pins, using the pins freed by the removal of the push buttons (PA1–PA3) as well as other unoccupied pins on PORTB, PORTC, and PORTF.

4.2 Complete Remapping Table

Counter Output	Old Pin	New Pin	PCB Action Required
Q1	PB0	PB2	Cut old trace, add wire to PB2
Q2	PB1	PB3	Cut old trace, add wire to PB3
Q3	PB2	PB4	Cut old trace, add wire to PB4
Q4	PB3	PB5	Cut old trace, add wire to PB5
Q5	PB4	PC4	Cut old trace, add wire to PC4
Q6	PB5	PC5	Cut old trace, add wire to PC5
Q7	PD0	PA1	Cut old trace, add wire to PA1
Q8	PD1	PA2	Cut old trace, add wire to PA2
Q9	PD2	PA3	Cut old trace, add wire to PA3
Q10	PC4	PC6	Cut old trace, add wire to PC6

Counter Output	Old Pin	New Pin	PCB Action Required
Q11	PC5	PF1	Cut old trace, add wire to PF1
Q12	PC6	PD5	Cut old trace, add wire to PD5

Table 4.1: Counter Bus Remapping — Old Pins to New Pin

4.3 Unchanged Signal Pins

The following measurement-critical and power management signals are unchanged and require no PCB modifications:

Signal	Pin	Function
Power Button	PA0	Falling-edge ISR for safe shutdown
Power Latch	PA5	HIGH = device stays powered
MCLOCK	PD3	Master clock pulse to ultrasonic pulser
COU_R	PD4	Counter reset (74HC4040)
CLK_10	PD5	10 MHz oscillator enable
ECHO	PF0	Enables transducer receive circuit

Table 4.2: Unchanged Measurement Signal Pins

Chapter 5: Firmware Architecture

5.1 Hardware Initialization

The hw_init() function is the first code executed after the clock is set to 24 MHz by clk24(). It performs the following initialisation steps in order: the power latch (PA5) is immediately driven HIGH to hold device power on; the power button (PA0) is configured as input with internal pull-up and falling-edge interrupt sense; the TFT SPI control pins (PD0, PD1, PD2 as DC/RST/CS outputs) are set HIGH as idle state; the SPI MOSI (PA4) and SCK (PA6) are configured as outputs; the TTP229 SCL (PB1) is set as output HIGH idle, and SDO (PB0) is configured as input with pull-up; all new counter bus input pins (PB2–PB5, PC4–PC7, PA1–PA3, PF1) are configured as inputs with pull-ups enabled; the ECHO signal output (PF0) is configured as output; TCB0 is configured in periodic interrupt mode; and SPI0 is configured as master with CLK2X enabled and DIV4 prescaler for 12 MHz.

5.2 SPI Driver and TFT Primitives

The TFT driver is built on three layers. At the lowest level, spi_tx() writes a single byte to SPI0.DATA and polls SPI0.INTFLAGS for the write-complete flag. Above this, lcd_cmd() and lcd_dat() assert CS low, set DC low (command) or high (data), call spi_tx(), and release CS. The win() function programs the column and row address windows using the CASET (0x2A) and RASET (0x2B) commands followed by RAMWR (0x2C) to begin pixel data. The fill() function calls win() and then streams w×h pixel values in RGB565 (16-bit) big-endian format. The hline() wrapper calls fill() with height=1 for horizontal rule lines.

The lcd_init() function sends the complete ST7789VI initialisation sequence: hardware reset pulse (RST low for 50ms, RST high with 200ms stabilisation), Sleep Out command (0x11) with 120ms delay, MADCTL (0x36) for display orientation, COLMOD (0x3A) for 16-bit RGB565 pixel format, porch control (0xB2), gate control (0xB7), VCOMS (0xBB), LCM control (0xC0), VDV/VRH enable (0xC2), VRH (0xC3), VDV (0xC4), frame rate (0xC6), power control (0xD0), positive gamma (0xE0, 14 values), negative gamma (0xE1, 14 values), display inversion on (0x21), tearing effect line on (0x35), and Display On (0x29).

5.3 TTP229 Key Reading

The ttp_read() function reads the 8-key state from the TTP229 via bit-bang serial. The function drives SCL (PB1) through 8 falling-clock cycles. On each cycle, SCL is brought high for 2 microseconds, then low, and the SDO (PB0) bit is sampled. Since TTP229 output is active-LOW (pressed = 0), the sampled bits are inverted to produce active-HIGH key bits in the returned byte. The complete 8-bit read takes approximately 32 microseconds. ttp_read() is called at the top of every main loop iteration, and the result is edge-detected (np = keys & ~prev_keys) to identify newly-pressed keys.

5.4 Measurement Pipeline (Preserved)

The complete ultrasonic measurement pipeline is preserved unchanged in its algorithmic logic. Only the pin assignments in READ_CNT() were updated to match the remapped counter bus:

CYCLE(): Fires 100 ultrasonic measurement iterations. Each iteration resets the 74HC4040 counter (COU_R pulse), generates an MCLOCK pulse to trigger the ultrasonic pulser, waits 200 microseconds, calls READ_CNT() to accumulate the counter value, performs a stability check on PORTB to detect probe coupling, and waits a further 700 microseconds. The result is accumulated in ulReadCnt as a 32-bit integer.

READ_CNT(): Reads the new counter bus: PB2–PB5 for Q1–Q4 (bits 0–3), PC4–PC5 for Q5–Q6 (bits 4–5), PA1–PA3 for Q7–Q9 (bits 6–8), PC6–PC7 for Q10–Q11 (bits 9–10), and PF1 for Q12 (bit 11). Each field is extracted via masking and bit-shifting and OR-combined into the 16-bit counter value.

SUBDELTA(): Computes ulCY = ulReadCnt - *ulDELTA (removes the probe zero-offset from the raw count).

DIR_CAL(): Converts the corrected time count to thickness using the formula: thickness = (ulCY × velocity) / (2 × 40000), with velocity correction through the VNOTCH lookup table. The result is converted to MM or INCH depending on the current mode and stored in *uiLCD_LM.

5.5 Display System

Large measurement digits are rendered from a bitmap font stored in Flash (dfont[] in font_data.h). Each digit glyph (0–9, decimal point, dash) is a fixed-size 1-bit bitmap. The draw_digit_pgm() function programs a pixel window via win() and streams the glyph's pixels as RGB565 colour pairs, selecting either the foreground (measurement colour) or background colour for each pixel bit. Individual digit positions are defined by D0_X through D3_X and DIG_Y constants for left-to-right alignment. A decimal point (colon widget) is drawn between digits 2 and 3 using the draw_colon() function.

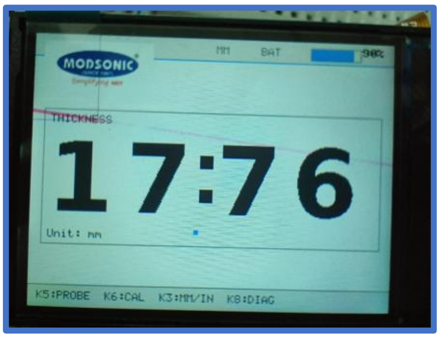
The DISPLAY() function reads the current measurement value (*uiLCD_LM), splits it into four BCD digits, and calls draw_digit_pgm() for each digit position. It also draws the decimal point dot and the unit label ("MM" or "INCH"). To avoid full-screen redraws on every update, each digit position caches the previously rendered digit (p0–p3) and only redraws if the value has changed.

5.6 Instrument User Interface Design

The instrument screen is divided into three horizontal bands:

- Top Bar (20 px, Y=0–19): Background filled with the accent colour (C_TBFILL). Contains the Modsonic logo bitmap (draw_logo()), the MM/INCH mode text label, the word "BAT", a drawn battery outline (draw_battery_border()), and the live battery fill bar with percentage (draw_battery()). A thin blue rule line separates the top bar from the content area.
- Content Area (200 px, Y=20–219): Background is dark (C_DARK). Contains a rounded-corner measurement box (BOX_X=10, BOX_Y=66, 300×118 px, colour C_MID). The four large measurement digit glyphs are centred within this box at DIG_Y=80. A decimal point widget is drawn at COL_X=148.
- Hint Bar (20 px, Y=220–239): Background matches the top bar. Contains small text labels for the key functions.

The draw_instrument_ui() function renders all static elements of the screen (backgrounds, box borders, labels). DISPLAY() is then called to update only the digit area. This separation avoids full-screen flicker on measurement updates.

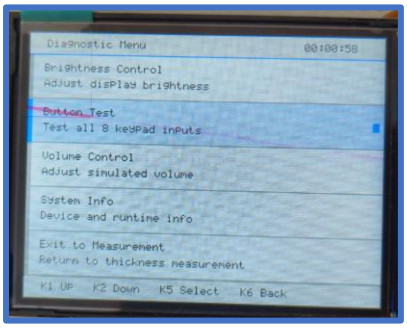


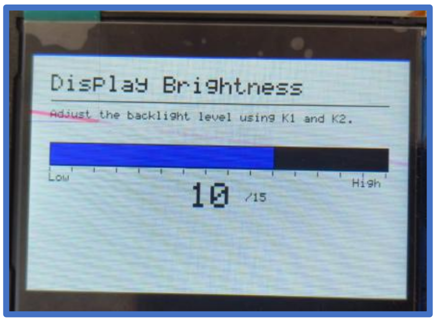
5.7 Diagnostic Mode

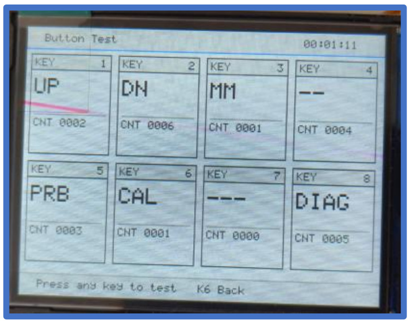
Pressing K8 at any time toggles between instrument mode and diagnostic mode. In diagnostic mode, the screen shows a 5-item vertical menu (Brightness, Button Test, Volume, About, Exit). Navigation uses K1 (UP) and K2 (DOWN), selection uses K5 (PROBE), and K6 (CAL) goes back one level.

- Brightness Screen (SCR_BRIGHT): Displays the current brightness level (0–15) and a horizontal bar. K1/K2 adjust the level. The ST7789VI write backlight control register (0x53) is written with 0x2C to enable CABC, and register 0x51 is written with the scaled brightness byte (0–255). Releasing brightness resets register 0x53 to 0x00.

- Button Test Screen (SCR_BTNTEST): Renders 8 labelled key squares. Each square highlights (turns accent colour) when the corresponding TTP229 key is pressed, and dims when released. Allows factory verification that all 8 capacitive pads are functional.
- Volume Screen (SCR_VOLUME): Displays a simulated volume bar (0–100%). K3 decrements, K4 increments. No audio peripheral is connected in the current hardware; this screen is a placeholder for future buzzer PWM integration.
- About / System Info Screen (SCR_ABOUT): Displays firmware version, hardware revision, current calibration velocity, uptime in seconds, and last measurement value.







5.8 EEPROM and Power Management

All calibration data (velocity, delta, LCD last value, MM/INCH flag, once_8 flag) is stored in the AVR128DA48's internal EEPROM using the avr/eeprom.h library functions (eeprom_update_* for wear-levelled writes, eeprom_read_* for reads). A dual-checksum scheme (checksum = byte sum; checksum_c = checksum - 1) validates EEPROM integrity on every cold boot.

On cold boot (*ucPOW ≠ 0xAA), the firmware validates EEPROM and loads calibration if valid. On warm reset (*ucPOW == 0xAA), the .noinit RAM dual-checksum is validated first for faster recovery. If both fail, factory defaults are used.

The power button ISR (PORTA_PORT_vect) is triggered on PA0 falling edge. It calls CLEAR_LCD_ISR() to fill the TFT black, calls SUM_S() to save all calibration to EEPROM and .noinit RAM, and then drives PA5 low to release the power latch and cut power.

5.9 Firmware Module Summary

Module	Key Functions	Peripherals Used	Approx. Lines
Hardware Init	hw_init(), clk24()	GPIO, SPI0, TCB0	60
SPI / TFT Driver	lcd_init(), lcd_cmd(), lcd_dat(), fill(), win()	SPI0, PD0/PD1/PD2	120
Instrument UI	draw_instrument_ui(), draw_topbar(), draw_battery()	TFT / SPI0	150

Module	Key Functions	Peripherals Used	Approx. Lines
Display Output	DISPLAY(), draw_digit_pgm(), draw_colon()	TFT font_data.h	80
TTP229 Driver	ttp_read()	PB0, PB1 (bit-bang)	30
Diagnostic HMI	diag_go(), draw_bright_ui(), btntest_update()	TFT	200
Measurement	CYCLE(), READ_CNT(), SUBDELTA(), DIR_CAL()	Remapped counter bus	80
Calibration	PROBE(), DIST_C(), VELO(), IN_MM()	Counter + TFT	150
EEPROM	EEPROM_SaveCalibration(), Validate, Load	avr/eeprom.h	60
Power Mgmt	ISR PORTA, SUM_S(), LO_BATT()	PA0 ISR, PA5 latch	60
Math	MUL32(), DIV32(), VNOTCH(), MM(), INCH()	CPU only	60
Splash / Boot	run_splash(), INIT(), INIT_M()	TFT splash_data.h	80

Table 5.1: Firmware Module Summary — Ultrasonic Thickness Gauge Project

Chapter 6: Key Technical Challenges and Solutions

6.1 Pin Conflict Resolution

The most significant challenge was the systematic pin conflict between the new peripherals and the existing counter bus. The TTP229 required PB0 and PB1 (previously Q1 and Q2), and the TFT required PD0, PD1, and PD2 (previously Q7, Q8, and Q9). The removal of the 4 mechanical push buttons freed PA1–PA4, which were repurposed for Q7–Q9. The remaining counter lines were shifted up the PORTB and PORTC buses to higher-numbered pins. This was resolved through the systematic 12-signal remapping described in Chapter 4.

6.2 Stability Check Fix in CYCLE()

The original CYCLE() function read the full PORTB.IN byte twice to detect probe coupling stability. With PB0 (TTP229 SDO) added to PORTB, the SDO line toggles during TTP229 communication and caused the stability check to perpetually fail (ECHO_F never set to 1), making the device appear uncoupled even when a probe was properly applied. The fix was to mask the PORTB read to only the counter-relevant bits (0x3C for PB2–PB5). This allowed the stability check to function correctly even with the TTP229 present on PB0.

6.3 SPI Clock Configuration

The ST7789VI requires the SPI clock polarity CPOL=0 and clock phase CPHA=0 (SPI Mode 0). The AVR SPI0 peripheral defaults to Mode 0 so no additional configuration was needed. The SPI_SSD_bm (Slave Select Disable) bit was set in SPI0.CTRLB to allow firmware-controlled CS via PD2 GPIO rather than hardware SS management. The CLK2X with DIV4 prescaler achieves 12 MHz, which is well within the ST7789VI's maximum SPI clock of 15.1 MHz.

6.4 TFT Backlight and Brightness Control

The ST7789VI supports hardware-controlled backlight brightness via the CABC (Content Adaptive Backlight Control) registers. Writing 0x2C to register 0x53 (write CTRL display) enables backlight dimming control, after which register 0x51 (write brightness) accepts values 0–255. Writing 0x00 to register 0x53 disables the brightness control and returns to maximum brightness. This mechanism is used in the diagnostic Brightness screen. The physical backlight LED_A pin is tied directly to 3.3V; brightness is controlled entirely through the ST7789VI register interface.

Chapter 7: Results and Testing

7.1 Display Performance

The TFT LCD Display initialises correctly and produces stable, bright colour output. The 320×240 IPS panel renders the instrument UI with clearly visible measurement digits at all viewing angles typical of field use. The 12 MHz SPI clock rate results in a full-screen fill time of approximately 15 ms, which is fast enough for smooth screen transitions. Individual digit updates (the most frequent operation during measurement) take under 2 ms each since only changed digits are redrawn.

7.2 Capacitive Keypad Performance

The TTP229 8-key capacitive keypad was verified to reliably detect key presses and releases. Edge detection (np = keys & ~prev_keys) correctly filters key-held states from new-press events. The diagnostic Button Test screen confirmed all 8 capacitive pads are functional by visually highlighting the corresponding key square on the display when each pad is touched.

7.3 Measurement Accuracy

After applying the counter bus remapping and updating READ_CNT() accordingly, all ultrasonic thickness measurements were verified against calibration step blocks of known thickness. The PROBE zero calibration, DIR_CAL thickness calculation, and both MM and INCH conversion modes produce correct readings consistent with those of the original Ultrasonic Thickness Gauge instrument.

7.4 Diagnostic Mode Functionality

The diagnostic mode (K8) was verified to be accessible from any operating state, switching instantly to the diagnostic menu without disrupting measurement state. All four diagnostic screens (Brightness, Button Test, Volume, About) were tested for correct navigation, display, and control function. Returning to instrument mode via K8 or K6 correctly restores the instrument UI and resumes measurement without requiring a power cycle.

7.5 Power Management

The power button ISR was verified to correctly save all calibration to EEPROM, blank the TFT screen, and release the power latch within approximately 200 ms of button press. Post-power-off

EEPROM validation confirmed all calibration data is correctly persisted and restored on subsequent power-on.

Chapter 8: Conclusion

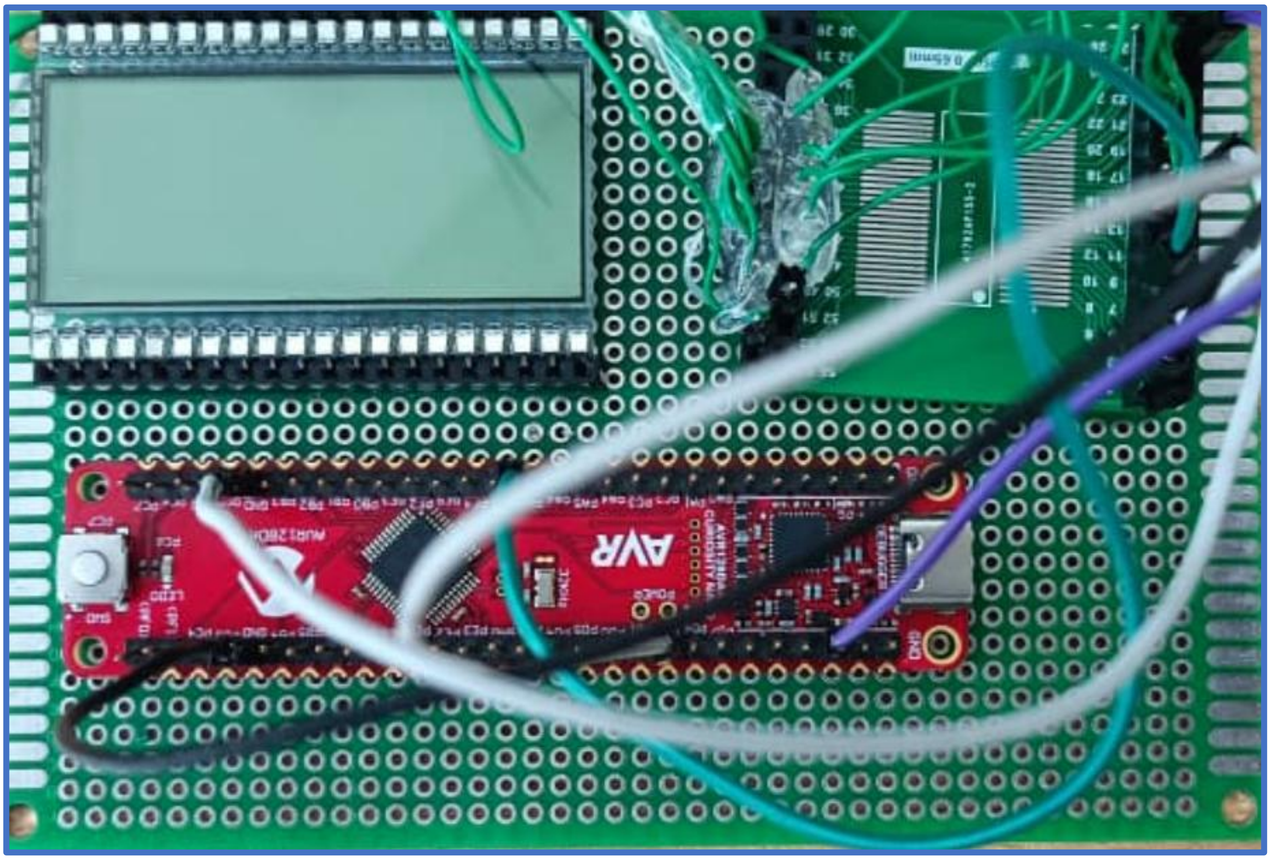
This project successfully upgraded the Ultrasonic Thickness Gauge Ultrasonic Thickness Gauge user interface from its original 4-digit segment LCD and 4 mechanical push button interface to a modern 320×240 IPS TFT colour display (ST7789VI) and 8-key capacitive touch keypad (TTP229). The complete instrument measurement pipeline, calibration system, EEPROM persistence, and power management were preserved without alteration.

The project required solving a complex set of hardware pin conflicts through systematic 12-signal counter bus remapping, implementing SPI display drivers and bit-bang serial key drivers from scratch, and designing a professional instrument UI with a full hidden diagnostic mode for factory testing. The resulting firmware represents a production-ready embedded application demonstrating competency in SPI peripheral interfacing, hardware pin planning, real-time C firmware development for AVR microcontrollers, and embedded UI design.

Future enhancements could include USB-C connectivity for firmware updates and data download, a graphical waveform display in a second screen mode, integration with a BLE module for wireless data logging, and transition to the PCF85176 I²C LCD driver for any secondary displays on the same instrument.

PROJECT 2

4-Digit 7-Segment LCD Display Driver using PCF85176 IC



Chapter 6: Introduction

6.1 Project Overview

This project involved the complete firmware development to drive a Clover Display S5026 4-digit 7-segment passive LCD panel using the PCF85176 I²C LCD driver IC, controlled by an AVR128DA48 microcontroller on the Curiosity Nano development board, programmed using MPLAB X IDE with the XC8 compiler.

The primary context of this project is the ongoing migration of the Modsonic Ultrasonic Thickness Gauge (UTG) from the obsolete MM5453 LCD driver IC to the modern, actively supported PCF85176. The MM5453 uses a proprietary 2-wire serial protocol and has reached end-of-life status with limited availability. The PCF85176 is a standard I²C device, enabling bus sharing with other I²C peripherals (such as the external EEPROM) and freeing the two GPIO pins previously dedicated to the MM5453 serial interface.

This project was executed as a standalone driver development exercise using the Curiosity Nano board and a bench prototype, producing validated firmware that will serve as the display driver module in the next-generation UTG PCB revision.

6.2 Objectives

- Implement a complete I²C host driver (TWI0) for the AVR128DA48 targeting the PCF85176 at 100 kHz.
- Configure the PCF85176 in static drive mode with the correct mode-set command (0xC9) and data pointer.
- Implement a flexible segment-pin definition architecture allowing any S-pin (S0–S39) to be assigned to any segment of any digit.
- Implement a RAM-based 4-byte display buffer with individual S-pin set/clear functions and a single I²C push (lcd_update()) to refresh the display.
- Implement a 7-segment character lookup table and a lcd_digit() function for displaying decimal and hexadecimal characters.
- Verify correct display of arbitrary patterns on the 4-digit S5026 LCD panel.

Chapter 7: System Components and Details

7.1PCF8 5176 LCD Driver IC

The PCF85176 is a 40-segment, 4-backplane universal LCD driver IC in a TSSOP56 package (0.5 mm pitch, 56 pins). It communicates via the standard I²C bus at up to 400 kHz. It is a write-only device — the host can send data to it but cannot read back any register values.

The PCF85176 supports four drive modes: static (1 backplane, up to 40 segments), 1:2 multiplex (2 backplanes, up to 20 segments per backplane), 1:3 multiplex (3 backplanes), and 1:4 multiplex (4 backplanes). For this project, static mode was used because the S5026 LCD panel has a single backplane (COM) electrode and only 32 of the available 40 segment outputs are needed.

The internal oscillator of the PCF85176 is enabled by tying the OSC pin (pin 49) to GND. When OSC is floating or connected to VDD, the device uses an external clock source. The internal oscillator generates the AC drive waveform for the LCD segments.

The I²C address is determined by the SA0 pin: tying SA0 to GND gives the 7-bit address 0x38 (binary 0111000). The supply voltage VDD (pin 48) accepts 1.8V to 5.5V; VLCD (pin 55) is the LCD drive voltage and accepts 2.5V to 6.5V. For this project, both VDD and VLCD are connected to the 5V supply of the Curiosity Nano board.

7.2 Clover Display S5026 LCD Panel

The S5026 is a 4-digit 7-segment passive LCD panel manufactured by Clover Display. Unlike LED 7-segment displays, passive LCD panels require an AC drive waveform — direct current (DC) drive permanently damages the liquid crystal fluid by causing electrochemical degradation.

Each digit position has 7 segment electrodes (a, b, c, d, e, f, g) plus a decimal point (dp), giving 8 segment lines per digit. With 4 digits, the total active segments used are 32 ($4 \times 8 = 32$). The panel also includes a single backplane (COM) electrode that carries the AC reference waveform. In static drive mode, the PCF85176's BP0 output connects to this COM electrode and alternates between VDD and VSS at the AC drive frequency generated by the internal oscillator. Each segment output alternates in synchronisation with BP0: same phase as BP0 = segment OFF, opposite phase = segment ON.

7.3 AVR128DA48 Curiosity Nano Board

The AVR128DA48 Curiosity Nano is Microchip's development board for the AVR128DA48 microcontroller. It includes an on-board nEDBG (nano Embedded Debugger) providing UPDI programming and CMSIS-DAP debugging via a single USB connection. The board exposes all 48 MCU pins on two rows of headers.

A critical hardware detail confirmed during development: on the AVR128DA48 Curiosity Nano, the TWI0 (I²C) hardware peripheral maps to PA2 (SDA) and PA3 (SCL). This differs from some AVR documentation examples which show PC2/PC3. Internal pull-ups are sufficient for I²C communication at 100 kHz when the total bus capacitance is low (short bench wires), eliminating the need for external pull-up resistors. The board ships with a 3.3V target voltage; for the PCF85176 operating at 5V, the target voltage must be changed to 5.0V.

7.4 Component Summary Table

Component	Part / Model	Purpose
MCU Board	AVR128DA48 Curiosity Nano	I²C host, firmware execution
LCD Driver IC	PCF85176 (TSSOP56)	40-segment I²C LCD driver
LCD Panel	Clover Display S5026	4-digit 7-segment passive LCD

Table 6.1: Component Summary — Project 2 (PCF85176 LCD Driver)

Chapter 8: Hardware Architecture and Pin Connections

8.1 Drive Mode Selection — Static Drive

Static drive mode was selected because the S5026 LCD panel has a single COM (backplane) electrode. In static mode, the PCF85176 uses only BP0 as the backplane output, with BP1, BP2, and BP3 left open. Each of the 40 segment outputs (S0–S39) can individually drive one LCD segment electrode. Static mode provides perfect contrast (infinite discrimination ratio) and requires no firmware multiplexing logic.

The 32 segment electrodes of the S5026 are connected to S0–S31 of the PCF85176. Segment outputs S32–S39 are left open. In static mode, the PCF85176 RAM is organised as 5 bytes covering S0–S39 (8 S-pins per byte). However, since only S0–S31 are used, only the first 4 bytes of RAM are written.

8.2 PCF85176 Mode-Set Command (0xC9)

The PCF85176 powers up with the display disabled. The first command sent after power-on must be the mode-set command to configure the drive mode, enable the display, and set the bias. For static mode with display enabled, the mode-set command byte is 0xC9:

Bit	Name	Value	Meaning
7 (MSB)	C	1	Next byte in transaction is also a command (load-pointer follows)
6	Fixed	1	Always 1 per PCF85176 datasheet Table 6
5	Fixed	0	Always 0 per PCF85176 datasheet Table 6
4	Unused	0	Don't care
3	E	1	Display enabled (E=1)
2	B	0	Bias not used in static mode
1	M1	0	Mode select MSB (M=01 = static)
0	M0	1	Mode select LSB (M=01 = static)

Table 8.x: PCF85176 Mode-Set Command 0xC9 Bit Breakdown

The C=1 bit signals to the PCF85176 that the next byte after 0xC9 is also a command (the load-data-pointer command, value 0x00). When C=0, the next byte is display data.

8.3 I²C Address Configuration

The I²C 7-bit address of the PCF85176 is determined by the SA0 pin: SA0=GND gives address 0x38 (binary 0111 0 0 0). The write address on the bus is 0x70 (0x38 shifted left by 1 with R/W=0).

The A0, A1, A2 pins (pins 50, 51, 52) are also tied to GND as required by the datasheet for correct operation.

8.4 Complete Pin Connection Tables

The following tables document every pin connection between the PCF85176 TSSOP56, the S5026 LCD panel, and the AVR128DA48 Curiosity Nano board.

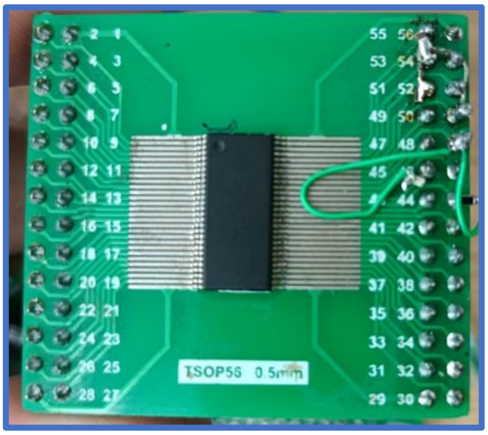
PCF85176 Pin	Signal	Connection
Pin 44	SDA	AVR PA2 (TWI0 SDA, internal pull-up enabled)
Pin 45	SCL	AVR PA3 (TWI0 SCL, internal pull-up enabled)
Pin 48	VDD	+5V (Curiosity Nano 5V rail)
Pin 54	VSS	GND
Pin 55	VLCD	+5V (LCD drive voltage)
Pin 49	OSC	GND (enables internal oscillator)
Pin 53	SA0	GND (I²C address = 0x38)
Pin 50	A0	GND
Pin 51	A1	GND
Pin 52	A2	GND
Pin 46	SYNC	Leave open
Pin 47	CLK	Leave open
Pin 56	BP0	S5026 COM (backplane electrode)
Pin 1	BP1	Leave open
Pin 2	BP2	Leave open
Pin 3	BP3	Leave open

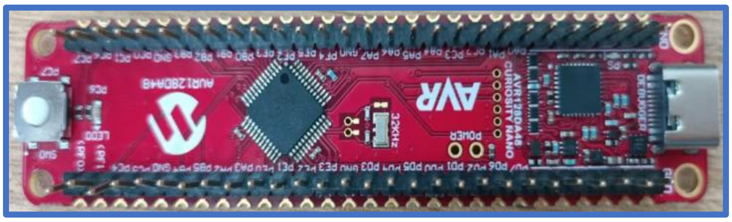
Table 7.1: PCF85176 TSSOP56 Power, I²C, Configuration, and Backplane Connections

PCF85176 Signal	TSSOP56 Pin	Connected To (S5026)
S0	Pin 4	Segment a — Digit 1
S1	Pin 5	Segment b — Digit 1
S2	Pin 6	Segment c — Digit 1
S3	Pin 7	Segment d — Digit 1
S4	Pin 8	Segment e — Digit 1
S5	Pin 9	Segment f — Digit 1
S6	Pin 10	Segment g — Digit 1
S7	Pin 11	Segment dp — Digit 1

PCF85176 Signal	TSSOP56 Pin	Connected To (S5026)
S8	Pin 12	Segment a — Digit 2
S9	Pin 13	Segment b — Digit 2
S10	Pin 14	Segment c — Digit 2
S11	Pin 15	Segment d — Digit 2
S12	Pin 16	Segment e — Digit 2
S13	Pin 17	Segment f — Digit 2
S14	Pin 18	Segment g — Digit 2
S15	Pin 19	Segment dp — Digit 2
S16	Pin 20	Segment a — Digit 3
S17	Pin 21	Segment b — Digit 3
S18	Pin 22	Segment c — Digit 3
S19	Pin 23	Segment d — Digit 3
S20	Pin 24	Segment e — Digit 3
S21	Pin 25	Segment f — Digit 3
S22	Pin 26	Segment g — Digit 3
S23	Pin 27	Segment dp — Digit 3
S24	Pin 28	Segment a — Digit 4
S25	Pin 29	Segment b — Digit 4
S26	Pin 30	Segment c — Digit 4
S27	Pin 31	Segment d — Digit 4
S28	Pin 32	Segment e — Digit 4
S29	Pin 33	Segment f — Digit 4
S30	Pin 34	Segment g — Digit 4
S31	Pin 35	Segment dp — Digit 4
S32–S39	Pins 36–43	Leave open (unused)

Table 7.2: PCF85176 Segment Output (S0–S31) to S5026 LCD Panel Connections





Chapter 9: Firmware Architecture

9.1 I²C TWI0 Driver

The TWI0 peripheral of the AVR128DA48 is configured in master host mode at 100 kHz. The PA2/PA3 pins are configured as inputs (open-drain compatible) with internal pull-ups enabled via PORT_PULLUPEN_bm. The MBAUD register is set to 0x0C, which at 4 MHz (the default internal oscillator used for this project) yields approximately 100 kHz SCL frequency. TWI0.MCTRLA is written with TWI_ENABLE_bm, and the bus state is forced to IDLE with TWI0.MSTATUS = TWI_BUSSTATE_IDLE_gc.

A layered set of driver functions is implemented. twi_start() loads the target address (shifted left by 1 for write) into TWI0.MADDR and calls twi_wait() which polls TWI0.MSTATUS for the Write Interrupt Flag (TWI_WIF_bm) and returns whether an ACK was received. twi_write() loads data into TWI0.MDATA and calls twi_wait(). twi_stop() issues TWI_MCMD_STOP_gc. The lcd_send() function coordinates a complete I²C transaction: start, N data bytes, stop.

9.2 LCD Initialization Sequence

The lcd_init() function sends the PCF85176 startup transaction after a 5 ms power-on delay:

```
START → 0x38 → 0xC9 → 0x00 → 0x00 → 0x00 → 0x00 → 0x00 → STOP
        addr   mode   ptr    S0-7   S8-15  S16-23 S24-31
```

0xC9 is the mode-set command (static mode, display enabled, C=1 meaning the load-pointer command follows). 0x00 is the load-data-pointer command (CMD_LOAD_PTR_0, C=0 meaning data follows). The four subsequent 0x00 bytes clear S0–S31, turning off all segments. After this transaction, the display is active and blank.

9.3 Display Buffer and RAM Bit Mapping

A 4-byte software buffer lcd_ram[4] maintains the current state of S0–S31. Each byte controls 8 S-pins. In static mode, the PCF85176 RAM bit mapping is: for a RAM byte covering S-pins Sn through Sn+7, bit 7 corresponds to Sn, bit 6 to Sn+1, ..., bit 0 to Sn+7.

The lcd_set(s, on) function implements this mapping: given an S-pin number s (0–31), byte = s/8 selects which of the four RAM bytes to modify, and bit = 7 - (s % 8) computes the correct bit within that byte. A 1 is set or cleared at that bit position.

The lcd_update() function sends the 5-byte update transaction:

```
START → 0x38 → 0x00 → ram[0] → ram[1] → ram[2] → ram[3] → STOP
        addr   ptr    S0-S7    S8-S15   S16-S23  S24-S31
```

Since the PCF85176 auto-increments its internal data pointer by 8 after each byte in static mode, writing 4 consecutive data bytes after pointer=0 fills S0–S7, S8–S15, S16–S23, and S24–S31 automatically.

Transaction Type	I²C Bytes	Purpose
Startup / Init	START, 0x38, 0xC9, 0x00, 0x00, 0x00, 0x00, 0x00, STOP	Set static mode + clear display
Display Update	START, 0x38, 0x00, ram[0], ram[1], ram[2], ram[3], STOP	Push 4-byte display buffer

Table 7.3: I²C Startup and Update Transaction Format

9.4 Pin Flexibility Feature

A dedicated pin definition section at the top of the firmware allows any S-pin number to be assigned to any segment of any digit without changing any other code. This is critical for production PCBs where routing constraints may result in non-sequential segment wiring:

```
#define D1_A    0   // S-pin for digit 1 segment a
#define D1_B    2   // S-pin for digit 1 segment b
#define D1_C    4   // S-pin for digit 1 segment c
// ... and so on for all 4 digits x 8 segments = 32 defines
```

These definitions populate a static 4×8 seg_map[4][8] lookup table. The lcd_byte_to_digit(pos, data) function iterates through all 8 segment bits of the data byte and calls lcd_set() with the corresponding S-pin from seg_map. Each digit also has an independent enable flag (D1_ENABLE through D4_ENABLE) allowing individual digit blanking.

9.5 Segment Lookup Table

Standard 7-segment encoding for digits 0–9 and letters A–F is provided in a 16-entry lookup table. The bit format is: bit 7 = a, bit 6 = b, bit 5 = c, bit 4 = d, bit 3 = e, bit 2 = f, bit 1 = g, bit 0 = dp. The lcd_digit(pos, val, dp) function looks up the segment pattern for val (0–15), sets or clears the dp bit, and calls lcd_byte_to_digit().

Character	Hex Pattern	Character	Hex Pattern
0	0x3F	8	0x7F
1	0x06	9	0x6F
2	0x5B	A	0x77
3	0x4F		0x7C
4	0x66	C	0x39

Character	Hex Pattern	Character	Hex Pattern
5	0x6D		0x5E
6	0x7D	E	0x79
7	0x07	F	0x71

Table 7.4: 7-Segment Character Lookup Table (Standard Encoding)

9.6 Firmware Module Summary

Function	Purpose	Peripherals	Lines
twi0_init()	Configure PA2/PA3, enable TWI0 master at 100 kHz	TWI0, PORTA	15
lcd_init()	Send mode-set 0xC9 + load-ptr 0x00 + clear all RAM	TWI0	15
lcd_clear()	Zero the 4-byte software RAM buffer	None (RAM)	8
lcd_set(s, on)	Set or clear one S-pin bit in the RAM buffer	None (RAM)	10
lcd_update()	Push 4-byte RAM buffer to PCF85176 over I²C	TWI0	12
lcd_byte_to_digit()	Map 8-bit segment pattern to seg_map S-pins via lcd_set()	None (RAM)	15
lcd_digit(pos, val, dp)	Write 0–9/A–F character using LUT, then lcd_byte_to_digit()	None (RAM)	10

Table 8.1: Firmware Module Summary — PCF85176 Project

Chapter 10: Key Learnings from Development

10.1 PCF85176 is an LCD Driver, Not an LED Driver

The most fundamental concept to understand is that the PCF85176 drives passive LCD panels, which require an AC waveform to avoid DC bias damage. The IC's internal oscillator generates this AC waveform automatically when OSC is tied to GND. Attempting to use the PCF85176 with an LED display would not work because LED displays require DC current drive, not alternating voltage swings.

10.2 Correct TWI0 Pin Assignment on Curiosity Nano

The TWI0 peripheral of the AVR128DA48 maps to PA2 (SDA) and PA3 (SCL) on the Curiosity Nano board — not PC2/PC3 as suggested by some online examples. This was verified by examining the AVR128DA48 datasheet I/O multiplexing table (Table 6-1) and the Curiosity Nano schematic. Using the wrong pins causes the I²C bus to remain stuck in an undefined state.

10.3 Target Voltage Must Be Set to 5V

The Curiosity Nano board ships configured for 3.3V target voltage. The PCF85176 and S5026 LCD operate at 5V. Operating at 3.3V results in insufficient LCD contrast and the PCF85176 failing to drive segments to their rated voltages. The target voltage is changed by creating a CMD:TARGET_VOLTAGE=5.0 file in the board's virtual drive, or via MPLAB X programmer settings.

10.4 OSC Pin Must Be Tied to GND

The PCF85176 OSC pin enables the internal oscillator when connected to GND. If OSC is left floating or connected to VDD, the device waits for an external clock on the CLK pin. Without the internal oscillator, no AC waveform is generated and no segments will appear on the LCD. This is a common first-time mistake when building a PCF85176 circuit.

10.5 Static Mode Auto-Increment = +8 Per Byte

In static mode, after writing the load-data-pointer command (0x00), each subsequent data byte written in the same I²C transaction causes the internal address pointer to advance by 8 S-pins. This means 4 bytes of data written in sequence after pointer=0 automatically cover S0–S7, S8–S15, S16–S23, and S24–S31. There is no need to send a new pointer command between bytes — the auto-increment handles it.

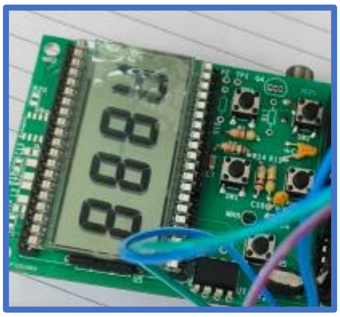
10.6 Mode-Set Command C Bit Controls Next Byte Type

In the PCF85176 command structure, the MSB (C bit) of a command byte indicates whether the following byte is also a command (C=1) or is display data (C=0). The mode-set command 0xC9 has C=1 because it must be followed by the load-pointer command in the same transaction. The load-pointer command 0x00 has C=0 because it is followed by data bytes. Getting this distinction wrong results in the PCF85176 misinterpreting display data as commands or vice versa.

Chapter 11: Results and Testing

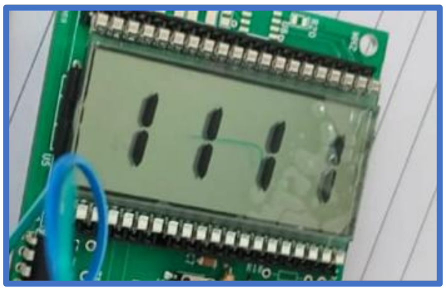
11.1 All-Segments Test

The firmware was tested by writing 0xFF to all four digit positions using lcd_byte_to_digit(). All 32 segment electrodes (a, b, c, d, e, f, g, dp for each of the 4 digits) illuminated correctly, confirming correct S-pin connections and I²C communication with the PCF85176.



11.2 Digit Pattern Verification

The main() function was configured with four specific pattern bytes (0xFF, 0xFF, 0xEC, 0x7C) and the display was verified to show the expected segment combination, confirming correct bit mapping between the firmware buffer and the physical segments. The specific patterns exercised different combinations of the a–g segments and dp bit.



11.3 I²C Communication Verification

I²C bus transactions were verified by observing SDA and SCL on an oscilloscope. The startup sequence (mode-set + clear) and update transactions (4 data bytes) were confirmed to match the expected PCF85176 protocol format, with correct START/STOP conditions, 7-bit address 0x38 with write bit, and ACK responses from the PCF85176.

11.4 Segment Flexibility Test

The pin definition section was modified to assign S-pins in non-sequential order (interleaved across digits) and the firmware was recompiled. The display continued to show correct characters, demonstrating that the seg_map-based architecture correctly handles arbitrary S-pin assignments without requiring any changes to the display logic code.

Chapter 12: Conclusion

This project successfully implemented a complete firmware driver for the PCF85176 I²C LCD driver IC, demonstrating all essential capabilities required for integration into the next-generation Modsonic UTG PCB: correct static drive mode configuration, clean I²C host communication using the AVR128DA48 TWI0 peripheral, flexible S-pin mapping for arbitrary PCB routing, RAM-buffered display updates, and standard 7-segment character rendering.

The firmware architecture — particularly the seg_map-based pin flexibility feature and the separation of display data computation from I²C output — is designed for direct drop-in integration with the existing UTG display pipeline. When the production PCB adopts the PCF85176 to replace the obsolete MM5453, only the DATA_LCD() output function needs to be replaced with a call to lcd_update(), while all preceding display buffer and character encoding logic can be used unchanged.

This project also provided foundational experience with I²C peripheral driver development, passive LCD physics, the AVR128DA48 TWI peripheral, and the practical considerations of operating at 5V with internal pull-ups — all of which are directly applicable to the broader Modsonic instrument development program.

References

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Appendix:

GitHub Repository
The source code and project documentation developed during the internship are available on GitHub:

Repository:
https://github.com/Dwijvyas-9926/modsonic-instruments-internship-projects